

# Siddharth Doshi

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## Contact Information

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## Profile

My research focuses on the study of the structure of neutron-rich nuclei far from stability, particularly through lifetime measurements,  $\beta$ - $\gamma$  spectroscopy, and the study of isomeric states. During my PhD, I have led the analysis of  $\beta$ - $\gamma$  spectroscopy and lifetime measurements in neutron-rich Hf and Ta isotopes, and have developed experience with ROOT, Radware and Python-based analysis workflows for  $\gamma$ -ray spectroscopy, coincidence analysis, peak fitting and half-life measurements. I am also analysing data from a storage-ring experiment at GSI, where I have developed Python-based tools for particle identification, frequency-spectrum analysis and multi-parameter fitting for mass and half-life measurements of neutron-rich nuclei. I also have a strong background in nuclear instrumentation, having contributed to experiments and data analysis at international research facilities with both stable and radioactive beams, and through hands-on work with HPGe detectors, plastic scintillators, electronics and data-acquisition systems at GSI. I have presented my work at major nuclear physics conferences in the UK and abroad, and have also contributed to the organisation of the UK Nuclear Physics Early Career Researcher Forum and the Institute of Physics (IOP) Nuclear Physics Conference 2026. Looking forward, I aim to develop a research-intensive career focused on proposing and leading experiments where I can continue to apply and expand my technical and analytical expertise.

## Education

**University of Brighton, United Kingdom**

**PhD, Nuclear Physics**

Doctoral Supervisors: Prof. Alison Bruce, Prof. Zsolt Podolyák

Thesis Title:  $\beta$ - $\gamma$  Spectroscopy of Isomeric and Ground States in Neutron-Rich Hafnium and Tantalum Isotopes

May 2023 - present  
Exp. viva: Sept./Oct. 2026

**REVA University, India**

**M.Sc., Physics, 1st Class with Distinction**

Dissertation Supervisors: Dr R. P. Singh, Prof. Siva Nasarayya Chari S.

Dissertation: Study of  $\gamma$ - $\gamma$  Angular Correlations of  $^{60}\text{Co}$  source

Nov. 2020 - Oct. 2022

**Veer Narmad South Gujarat University, India**

**Dual B.Sc. Degrees, Physics and Mathematics, 1st Class with Distinction**

B.Sc. Physics: June 2016 - April 2019

B.Sc. Mathematics: June 2019 - April 2020

June 2016 - April 2020

## Publications

**S. Doshi**, P.M. Walker, A.M. Bruce, S. Pascu, Zs. Podolyák et al., Evidence for an isomeric  $\pi_{\frac{1}{2}}^{+}[411]$  state in  $^{183}\text{Ta}$ , *Eur. Phys. J. A* **62** (2026), 17

DOI: 10.1140/epja/s10050-025-01776-2

D. Freire-Fernández, ..., **S. Doshi** et al., Precision masses of neutron-rich platinum and gold nuclei reveal enhanced N=126 shell strength below doubly-magic  $^{208}\text{Pb}$ , *arXiv:2607.10894 [nucl-ex]* (2026)

DOI: 10.48550/arXiv.2607.10894

J. M. Yap, ..., **S. Doshi** et al., First Direct Mass Measurement of  $^{184}\text{Ta}$  and Surrounding Nuclei, *Atoms* **14**(8) (2026), 63

DOI: 10.3390/atoms14080063

- J. Ruotsalainen, ..., **S. Doshi** et al., Probing the quantum phase transition near  $N \approx 60$  via mass measurements of technetium isotopes, *Phys. Rev. C* **112** (2025) 6, 064302  
DOI: 10.1103/PhysRevC.112.064302
- S. Pascu, ..., **S. Doshi** et al., Collective excitations in  $^{150}\text{Gd}$ , *Phys. Rev. C* **111** (2025) 3, 034302  
DOI: 10.1103/PhysRevC.111.034302
- S. Pascu, ..., **S. Doshi** et al., Increasing Octupole Collectivity across the  $Z = 64$  Isotopic Chain:  $B(E3)$  Values in  $^{150}\text{Gd}$ , *Phys. Rev. Lett.* **134** (2025) 9, 092501  
DOI: 10.1103/PhysRevLett.134.092501
- S. Doshi**, S. Pascu, P.M. Walker, A.M. Bruce, Zs. Podolyák et al., Yield of neutron-rich  $^{183,184}\text{Hf}$  isotopes from KISS, *RIKEN Accelerator Progress Report* **57** (2024).
- S. Doshi**, Archit Gupta et al., Understanding of interaction of  $\gamma$ -rays with matter: Attenuation coefficient measurements, *Proceedings of the DAE Symp. on Nucl. Phys.* **66** (2022).

## Research Experience

### PhD Research Student (May 2023 - Present)

Throughout my PhD, I have contributed to nuclear-structure experiments and data analysis involving laser spectroscopy, decay spectroscopy, mass spectrometry,  $\gamma$ -ray spectroscopy, lifetime measurements, and studies using both stable and radioactive beams at:

- **CERN/ISOLDE, Switzerland:**
  - Laser and decay spectroscopy and mass spectrometry of neutron-rich mercury isotopes south-east of  $^{208}\text{Pb}$  (IS756, 21-31 Oct. 2024).
  - Transition probabilities of low-lying excited states in  $^{210}\text{Po}$  and  $^{210}\text{Pb}$  (IS720, 04-06 Nov. 2025).
- **IFIN-HH, Romania:**
  - Study of the  $^{150}\text{Tb}$  decay (29 Jan.-14 Feb. 2024).
  - $\gamma$ -spectroscopy of  $^{172}\text{Er}$  (07-16 Nov. 2025).
- **University of Jyväskylä, Finland:**
  - Decay properties of a new isomeric state in  $^{128}\text{In}$  (I291, 08-11 Mar. 2024).
  - Decay studies of  $A = 115$  nuclei for nuclear astrophysics (I294, 11-20 Mar. 2024).
- **GSI, Germany:**
  - Bromine Sandbanks experiment (G-22-00115, 26 Feb.-04 Mar. 2025).
  - Half-life measurements in the neutron-rich  $N \approx 116$  Hf region, nuclear two-photon decay, and bound-state pair conversion (G22\_E00018 and G24\_E00247, 31 Mar.-09 Apr. 2025).

My main PhD work is based on data from the KEK Isotope Separation System Facility (KISS), RIKEN, where neutron-rich  $^{183,184}\text{Hf}$  and  $^{183,184}\text{Ta}$  isotopes were produced through multi-nucleon transfer reactions and studied using  $\beta$ - $\gamma$  spectroscopy. The analysis focused on:

- Using GRAZING calculations and experimental data to estimate production yields of neutron-rich Hf and Ta isotopes from multi-nucleon transfer reactions.
- Identifying isomeric states, delayed  $\gamma$ -ray transitions, and associated decay structures.
- Interpreting the observed low-lying states using Nilsson configurations and systematics of neighbouring nuclei.
- Developing Radware, ROOT and Python-based workflows for  $\gamma$ -ray spectroscopy,  $\beta$ - $\gamma$  timing correlations, coincidence analysis, peak fitting, and half-life measurements.
- Preparing results for publication using  $\text{\LaTeX}$  and Python.

### Experimental Storage Ring Data Analysis

In addition to the above-mentioned work, I am also analysing data from a storage-ring experiment at GSI. The experiment is motivated by mass and half-life measurements in the neutron-rich  $N = 116$  Hf

region. Through this analysis, my work includes:

- Investigating isomers in neutron-rich Hf isotopes produced in fragmentation reactions.
- Developing Python tools for reading and analysing event-by-event data, particle identification, frequency-spectrum analysis and multi-parameter fitting.

### **Long-term Attachment with the HISPEC/DESPEC Group at GSI (Jan. 2025 - June 2025)**

I spent six months with the HISPEC/DESPEC group at GSI, working on DEGAS germanium detectors and BPlast position-sensitive plastic scintillator detectors with multi-SiPM readout. My work there involved:

- Contributing to the assembly and cryogenic setup of a new DEGAS detector station.
- Preparing cold boards, preamplifiers, and high-voltage boards for the DEGAS detectors.
- Detector handling, stacking, post-installation testing, and calibration using Cubix.
- Working with both analogue and digital electronics DAQ systems.
- Assisting with the preparation of a new implantation stack and BPlast detectors.
- Contributing to the planning and running of experiments.

### ***Other Relevant Experience***

- Provided short-term academic tutoring in physics and mathematics during my Master's studies.
- Taught and demonstrated laboratory experiments to high-school and undergraduate students during my Master's studies.
- Took a leading role in the organisation of the UK Nuclear Physics Early Career Researcher Forum in July 2026, including establishing and promoting programme sessions in which leading early career nuclear physicists and industry representatives presented their experiences.
- Contributed to the organisational work for the IOP Nuclear Physics Conference 2026.

### ***Presentation of Research***

These presentations focused primarily on my PhD work on isomeric states in  $^{183,184}\text{Ta}$ :

- April 2024 (Poster): IOP Joint APP, HEPP and NP Conference, Liverpool, UK.
- August 2024 (Talk): STFC Nuclear Physics Summer School, Durham, UK.
- April 2025 (Talk): IOP Nuclear Physics Conference, Manchester, UK.
- September 2025 (Talk): European Nuclear Physics Conference, Caen, France.
- February 2026 (Invited talk): ILIMA Open Meeting during the NUSTAR Annual Meeting, GSI, Germany.
- April 2026 (Featured talk): IOP Nuclear Physics Conference, Brighton, UK.

### ***Relevant Skills and Experience***

- **Nuclear spectroscopy and data analysis:** Experience in  $\gamma$ -ray spectroscopy,  $\beta$ - $\gamma$  timing correlations, coincidence analysis, peak fitting, half-life measurements, production-yield estimation, and interpretation of isomeric and decay structures, including  $K$ -isomers, Nilsson configurations, transition strengths and selection rules.
- **Programming and analysis tools:** Experience with ROOT, Radware, Python, C++, L<sup>A</sup>T<sub>E</sub>X, Microsoft Office, Unix/Linux and Windows operating systems. Developed analysis workflows for event-by-event spectroscopy data, particle identification, frequency-spectrum analysis, and multi-parameter fitting.
- **Detector systems and instrumentation:** Hands-on experience with HPGe detectors, plastic scintillator detectors with SiPM readout, analogue and digital electronics, detector calibration, data-

acquisition systems, and experimental setup work.

- **Experimental facilities:** Working knowledge of nuclear-physics facilities and experimental setups, including ISOLDE, ROSPHERE, IGISOL, KISS, ESR and HISPEC/DESPEC.
- **Radiation safety and experimental work:** Experience handling radioactive materials, working in radiation-controlled areas, and following experimental safety procedures.
- **Simulation and reaction calculations:** Basic working familiarity with Geant4 simulation concepts and experience using GRAZING calculations for multi-nucleon transfer reaction studies.
- **Communication and collaboration:** Experience presenting research at national and international conferences, and working independently and within large collaborations.
- **Organisational skills:** Contributed to the organisation of Early Career Researcher Forum activities and the IOP Nuclear Physics Conference 2026, including programme planning, session coordination and communication with speakers.